KONSTANTINOS VAMVAKAS

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EDUCATION

National Technical University of Athens (NTUA) **Expected 2028**
Diploma in Electrical and Computer Engineering (Integrated MEng)
GPA: 9.1/10 (excellent)

Arsakeia Schools, Psychiko 2022
High School Diploma
Grade: 19.5/20 (Excellent)

TECHNICAL SKILLS

- ***Programming:*** *C/C++, MATLAB*
- ***Electronics and Embedded Systems:*** *PCB design (KiCad), circuit simulation (LTspice), microcontrollers*
- ***Analysis and Control:*** *System modeling, digital signal processing (DSP), automatic control*
- ***Laboratory Skills:*** *Soldering, circuit debugging, electronic testing*

RESEARCH AND ENGINEERING EXPERIENCE

Student Research Engineer – Electric Racing Boat Team OCEANOS

School of Naval Architecture & Marine Engineering, NTUA 2024 - Present

- Designed and developed **custom PCBs** for power distribution, telemetry, control, motor pre-charge, and sensor interface systems.
- Achieved first-pass functional hardware with zero post-fabrication.
- Responsible for digital signal processing (**DSP**), including modeling and filter design in MATLAB and implementation on embedded microcontrollers.
- Developed and deployed C/C++ code for real-time control, signal processing, and system integration on onboard microcontrollers.
- Performed system-level modeling, simulation, and performance evaluation using MATLAB.
- Collaborated with Naval Architects and Mechanical Engineers to ensure reliable integration of electronics systems under racing conditions.

Undergraduate Researcher - Circuits & Systems Group
Prof. Paul P. Sotiriadis, NTUA 2025 - Present

- Analyze and design **nonlinear oscillator circuits** and dynamical systems using LTspice.
- Develop MATLAB-based models for simulation, stability analysis, and performance evaluation of oscillatory systems.
- Investigate control strategies for oscillators, including the application of **fuzzy control techniques**.
- Implement and evaluate control algorithms with potential deployment on embedded platforms.
- Collaborate with faculty on research-oriented problems in control systems and circuit design.

ACADEMIC PROJECTS

AM Radio Receiver Design Project

- Designed, built, and tested a complete AM radio receiver using discrete components.
- Performed circuit design, component selection, and implementation.
- Conducted debugging and validation using laboratory measurement equipment.

ADDITIONAL INFORMATION

Work experience: English Exams Invigilator – Esolnet Hellas (LRN Exams) 2023 - Present
Languages: *Greek (Native), English (C2), French (B2)*